



TEST REPORT

Report Number..... : ZKT-2110295824E

Date of Test..... Oct.29, 2021 to Nov.5, 2021

Date of issue..... : Nov.5, 2021

Total number of pages..... 29

Test Result : PASS

Testing Laboratory..... : Shenzhen ZKT Technology Co., Ltd.

Address 1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China

Applicant's name : Shenzhen Noblefull Technology Co., Ltd.

Address Honghai Building 2001, Hongxing Community, Songgang Street, Shenzhen, Guangdong, China.

Manufacturer's name : Shenzhen Noblefull Technology Co., Ltd.

Address Honghai Building 2001, Hongxing Community, Songgang Street, Shenzhen, Guangdong, China.

Test specification:

Standard..... EN 55032:2015+A11:2020, EN 55035:2017+A11:2020
EN IEC 61000-3-2:2019/A1:2021, EN 61000-3-3:2013+A1:2019
EN 61000-4-2:2009, EN IEC 61000-4-3:2020,
EN 61000-4-4:2012, EN 61000-4-5:2014+A1:2017,
EN 61000-4-6:2014, EN 61000-4-8:2010,
EN IEC 61000-4-11:2020

Test procedure..... : /

Non-standard test method : N/A

Test Report Form No..... : TRF-EL-144_V0

Test Report Form(s) Originator..... : ZKT Testing

Master TRF : Dated: 2020-01-06

This device described above has been tested by ZKT, and the test results show that the equipment under test (EUT) is in compliance with the 2014/30/EU Directive requirements. And it is applicable only to the tested sample identified in the report.

This report shall not be reproduced except in full, without the written approval of ZKT, this document may be altered or revised by ZKT, personal only, and shall be noted in the revision of the document.

Product name..... : USB AM to Type C Cable

Trademark : N/A

Model/Type reference..... NPF04A0020C001
NPF04A0020C002,NPF04A0030C001,NPF04A0030C002

Ratings..... : Input: DC 20V



Testing procedure and testing location:

Testing Laboratory.....: **Shenzhen ZKT Technology Co., Ltd.**

Address.....: 1/F, No. 101, Building B, No. 6, Tangwei Community
Industrial Avenue, Fuhai Street, Bao'an District,
Shenzhen, China

Tested by (name + signature).....: **Alen He**

Reviewer (name + signature).....: **Joe Liu**

Approved (name + signature).....: **Lake Xie**





TABLE OF CONTENT

	Page
1. VERSION	5
2. GENERAL INFORMATION	6
2.1 Description of Device (EUT)	6
2.2 Tested System Details	6
2.3 Test Facility	6
2.4 MEASUREMENT UNCERTAINTY	6
2.5 Test Instrument Used	7
3. CONDUCTED EMISSIONS	9
3.1 Block Diagram Of Test Setup	9
3.2 Limit	9
3.3 Test procedure	9
3.4 Test Result	9
4. RADIATED EMISSIONS TEST	10
4.1 Block Diagram Of Test Setup	10
4.2 Limits	10
4.3 Test Procedure	11
4.4 Test Results	12
5. HARMONIC CURRENT EMISSION TEST	14
5.1 Block Diagram of Test Setup	14
5.2 Test Standard	14
5.3 Operating Condition of EUT	14
5.3.1 Setup the EUT as shown in Section 6.1	14
5.3.2 Turn on the power of all equipment	14
5.3.3 Let the EUT work in test mode and test it	14
5.4 Test Procedure	14
5.5 Test Results	14
6. VOLTAGE FLUCTUATIONS & FLICKER TEST	15
6.1 Block Diagram of Test Setup	15
6.2 Test Standard	15
6.3 Operating Condition of EUT	15
6.4 Test Procedure	15
6.5 Test Results	15
7. IMMUNITY TEST OF GENERAL THE PERFORMANCE CRITERIA	16
8. ELECTROSTATIC DISCHARGE (ESD)	17
8.1 Test Specification	17
8.2 Block Diagram of Test Setup	17
8.3 Test Procedure	17
8.4 Test Results	18
9. CONTINUOUS RF ELECTROMAGNETIC FIELD DISTURBANCES(RS)	19
9.1 Test Specification	19
9.2 Block Diagram of Test Setup	19
9.3 Test Procedure	20



9.4 Test Results.....	20
10. ELECTRICAL FAST TRANSIENTS/BURST (EFT).....	21
10.1 Test Specification.....	21
10.2 Block Diagram of EUT Test Setup.....	21
10.3 Test Procedure.....	21
10.4 Test Results.....	21
11. SURGES IMMUNITY TEST.....	22
11.1 Test Specification.....	22
11.2 Block Diagram of EUT Test Setup.....	22
11.3 Test Procedure.....	22
11.4 Test Result.....	22
12. CONTINUOUS INDUCED RF DISTURBANCES (CS).....	23
12.1 Test Specification.....	23
12.2 Block Diagram of EUT Test Setup.....	23
12.3 Test Procedure.....	23
12.4 Test Result.....	23
13. MAGNETIC FIELD IMMUNITY TEST.....	24
13.1 Block Diagram of Test Setup.....	24
13.2 Test Standard.....	24
13.3 Severity Levels and Performance Criterion.....	24
13.3.1 Severity level.....	24
13.3.2 Performance criterion: B.....	24
13.4 EUT Configuration on Test.....	25
13.5 Operating Condition of EUT.....	25
13.6 Test Procedure.....	25
13.7 Test Results.....	25
14. VOLTAGE DIPS AND INTERRUPTIONS (DIPS).....	26
14.1 Test Specification.....	26
14.2 Block Diagram of EUT Test Setup.....	26
14.3 Test Procedure.....	26
14.4 Test Result.....	26
15. EUT PHOTOGRAPHS.....	27
16. EUT TEST PHOTOGRAPHS.....	29



1. VERSION

Report No.	Version	Description	Approved
ZKT-2110295824E	Rev.01	Initial issue of report	Nov. 5, 2021



2. GENERAL INFORMATION

2.1 Description of Device (EUT)

EUT : USB AM to Type C Cable

Trademark : N/A

Model Number : NPF04A0020C001
NPF04A0020C002,NPF04A0030C001,NPF04A0030C002

Model Difference : Only for different model name

Power Supply : Input:DC 20V

The highest frequency of the internal sources of the EUT is (less than 108)MHz:

- ☒ less than 108 MHz, the measurement shall only be made up to 1 GHz.
- ☐ between 108 MHz and 500 MHz, the measurement shall only be made up to 2 GHz.
- ☐ between 500 MHz and 1 GHz, the measurement shall only be made up to 5 GHz.
- ☐ above 1 GHz, the measurement shall be made up to 5 times the highest frequency or 6 GHz, whichever is less.

Note: N/A

2.2 Tested System Details

None.

2.3 Test Facility

Site Description

Name of Firm : Shenzhen ZKT Technology Co., Ltd.

Site Location : 1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China

2.4 MEASUREMENT UNCERTAINTY

Where relevant, the following measurement uncertainty levels have been estimated for tests performed on the Product as specified in CISPR 16-4-2. This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

Test item	Value (dB)
Conducted Emission (150K-30MHZ)	3.20
Radiated disturbance30MHz-1000MHz	4.80
Radiated disturbance1000MHz-6000MHz	5.10



2.5 Test Instrument Used

Conducted emissions Test

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	LISN	R&S	ENV216	101471	Sep. 22, 2021	Sep. 21, 2022
2	LISN	CYBERTEK	EM5040A	E1850400149	Sep. 22, 2021	Sep. 21, 2022
3	Test Cable	N/A	C01	N/A	Sep. 22, 2021	Sep. 21, 2022
4	Test Cable	N/A	C02	N/A	Sep. 22, 2021	Sep. 21, 2022
5	EMI Test Receiver	R&S	ESRP3	101946	Sep. 22, 2021	Sep. 21, 2022
6	Absorbing Clamp	DZ	ZN23201	N/A	Sep. 22, 2021	Sep. 21, 2022

Radiated emissions Test (966 chamber)

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	Bilog Antenna	Schwarzbeck	VULB9168	00877	Sep. 22, 2021	Sep. 21, 2022
2	Loop Antenna	SCHWARZBECK	FMZB1519B	014	Sep. 22, 2021	Sep. 21, 2022
3	Test Cable	N/A	R-01	N/A	Sep. 22, 2021	Sep. 21, 2022
4	Test Cable	N/A	R-02	N/A	Sep. 22, 2021	Sep. 21, 2022
5	EMI Test Receiver	R&S	ESCI7	101169	Sep. 22, 2021	Sep. 21, 2022
6	Antenna Mast	EM	SC100_1	N/A	N/A	N/A
7	Turn Table	EM	SC100	N/A	N/A	N/A
8	Spectrum Analyzer	KEYSIGHT	9020A	MY55370835	Sep. 22, 2021	Sep. 21, 2022
9	Horn Antenna (1GHz-18GHz)	Schwarzbeck	BBHA9120D	1541	Sep. 22, 2021	Sep. 21, 2022
10	Horn Antenna (18GHz-40GHz)	A.H. System	SAS-574	588	Sep. 22, 2021	Sep. 21, 2022
11	Amplifier (30-1000MHz)	EM Electronics	EM330 Amplifier	N/A	Sep. 22, 2021	Sep. 21, 2022
12	Amplifier (1GHz-40GHz)	全聚达	DLE-161	097	Sep. 22, 2021	Sep. 21, 2022

Harmonic / Flicker Test

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	Harmonic & Flicker	LAPLACE INSTRUMENTS	C2000A	311370	Sep. 22, 2021	Sep. 21, 2022
2	AC Power Source	LAPLACE INSTRUMENTS	C2000A	311370	Sep. 22, 2021	Sep. 21, 2022

Electrostatic discharge Test

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	ESD TEST GENERATOR	HTEC	HESD16	N/A	Sep. 22, 2021	Sep. 21, 2022



Continuous RF electromagnetic field disturbances Test (SMQ --- site)

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	Signal Generator	R&S	SMT 06	832080/007	Sep. 22, 2021	Sep. 21, 2022
2	Log-Bicon Antenna	Schwarzbeck	VULB9161	4022	Sep. 22, 2021	Sep. 21, 2022
3	Power Amplifier	AR	150W1000M 1	320946	Sep. 22, 2021	Sep. 21, 2022
4	Microwave Horn Antenna	AR	AT4002A	321467	Sep. 22, 2021	Sep. 21, 2022
5	Power Amplifier	AR	25S1G4A	308598	Sep. 22, 2021	Sep. 21, 2022

EFT and Surge and Voltage dips and interruptions Test

Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	Surge Generator	HTEC	HCOMPAC T5	202501	Sep. 22, 2021	Sep. 21, 2022
2	DIPS Generator	HTEC	HV1P16T	202101	Sep. 22, 2021	Sep. 21, 2022
3	EFT/B Generator	HTEC	HCOMPAC T5	202501	Sep. 22, 2021	Sep. 21, 2022
4	EFT/B Clamp	HTEC	H3C	N/A	Sep. 22, 2021	Sep. 21, 2022

For Magnetic Field Immunity Test

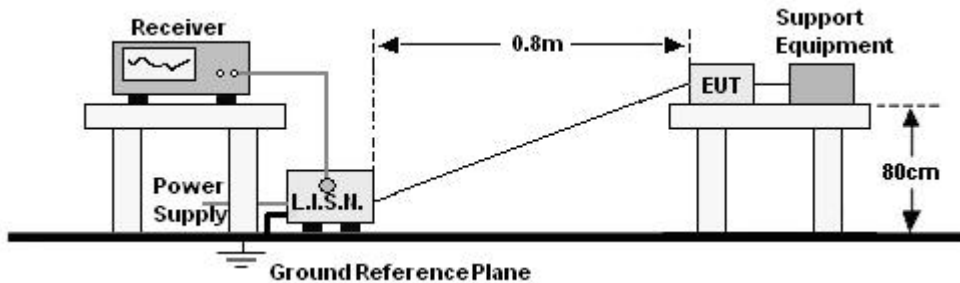
Item	Kind of Equipment	Manufacturer	Type No.	Serial No.	Last calibration	Calibrated until
1	Generator	HTEC	HFMG 100	202602	Sep. 22, 2021	Sep. 21, 2022



3. CONDUCTED EMISSIONS

3.1 Block Diagram Of Test Setup

For mains ports:



3.2 Limit

Limits for Conducted emissions at the mains ports of Class B MME

Frequency range (MHz)	Limits dB(μV)	
	Quasi-peak	Average
0,15 to 0,50	66 to 56*	56 to 46*
0,50 to 5	56	46
5 to 30	60	50

Notes: 1. *Decreasing linearly with logarithm of frequency.
2. The lower limit shall apply at the transition frequencies.

3.3 Test procedure

For mains ports:

- The Product was placed on a nonconductive table 0.8 m above the horizontal ground reference plane, and 0.4 m from the vertical ground reference plane, and connected to the main through Line Impedance Stability Network (L.I.S.N).
- The RBW of the receiver was set at 9 kHz in 150 kHz ~ 30MHz with Peak and AVG detector in Max Hold mode. Run the receiver's pre-scan to record the maximum disturbance generated from Product in all power lines in the full band.
- For each frequency whose maximum record was higher or close to limit, measure its QP and AVG values and record.

3.4 Test Result

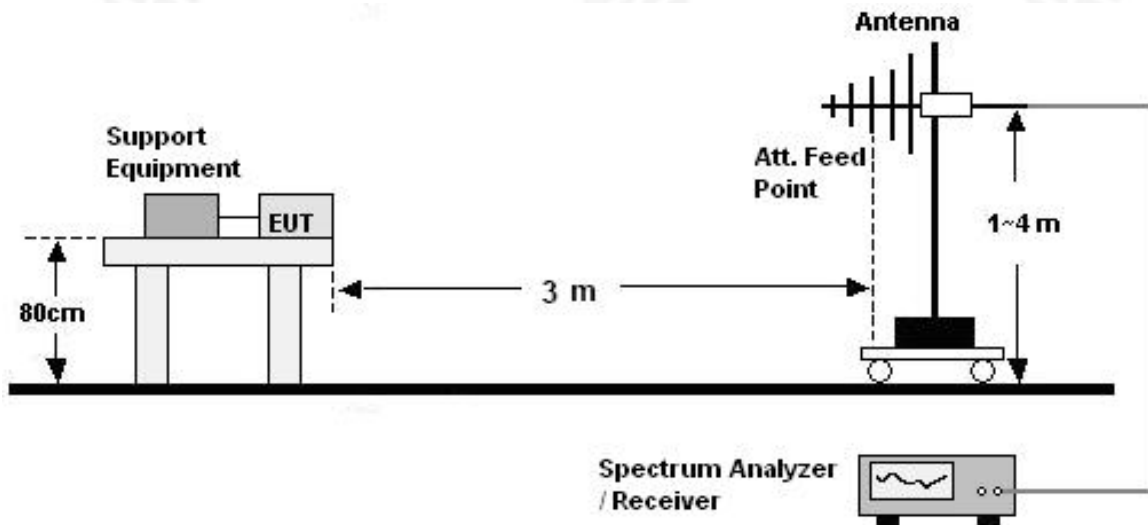
The EUT is powered by DC only. The test items is not applicable.



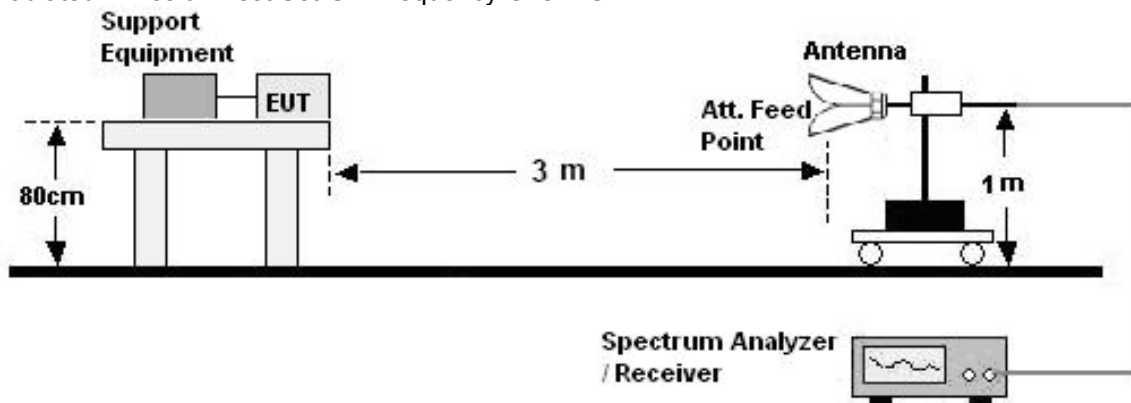
4. RADIATED EMISSIONS TEST

4.1 Block Diagram Of Test Setup

(A) Radiated Emission Test Set-UP Frequency 30MHz-1GHz



(B) Radiated Emission Test Set-UP Frequency Over 1GHz



4.2 Limits

Limits for radiated disturbance of Class B MME

Frequency (MHz)	Quasi-peak limits at 3m dB(μ V/m)
30-230	40
230-1000	47

FREQUENCY (MHz)	Class B (at 3m) dB μ V/m	
	Peak	Avg
1000-3000	70	50
3000-6000	74	54



4.3 Test Procedure

30MHz ~ 1GHz:

- a. The Product was placed on the nonconductive turntable 0.8 m above the ground in a semi anechoic chamber.
- b. Set the spectrum analyzer/receiver in Peak detector, Max Hold mode, and 120 kHz RBW. Record the maximum field strength of all the pre-scan process in the full band when the antenna is varied between 1~4 m in both horizontal and vertical, and the turntable is rotated from 0 to 360 degrees.
- c. For each frequency whose maximum record was higher or close to limit, measure its QP value: vary the antenna's height and rotate the turntable from 0 to 360 degrees to find the height and degree where Product radiated the maximum emission, then set the test frequency analyzer/receiver to QP Detector and specified bandwidth with Maximum Hold Mode, and record the maximum value.

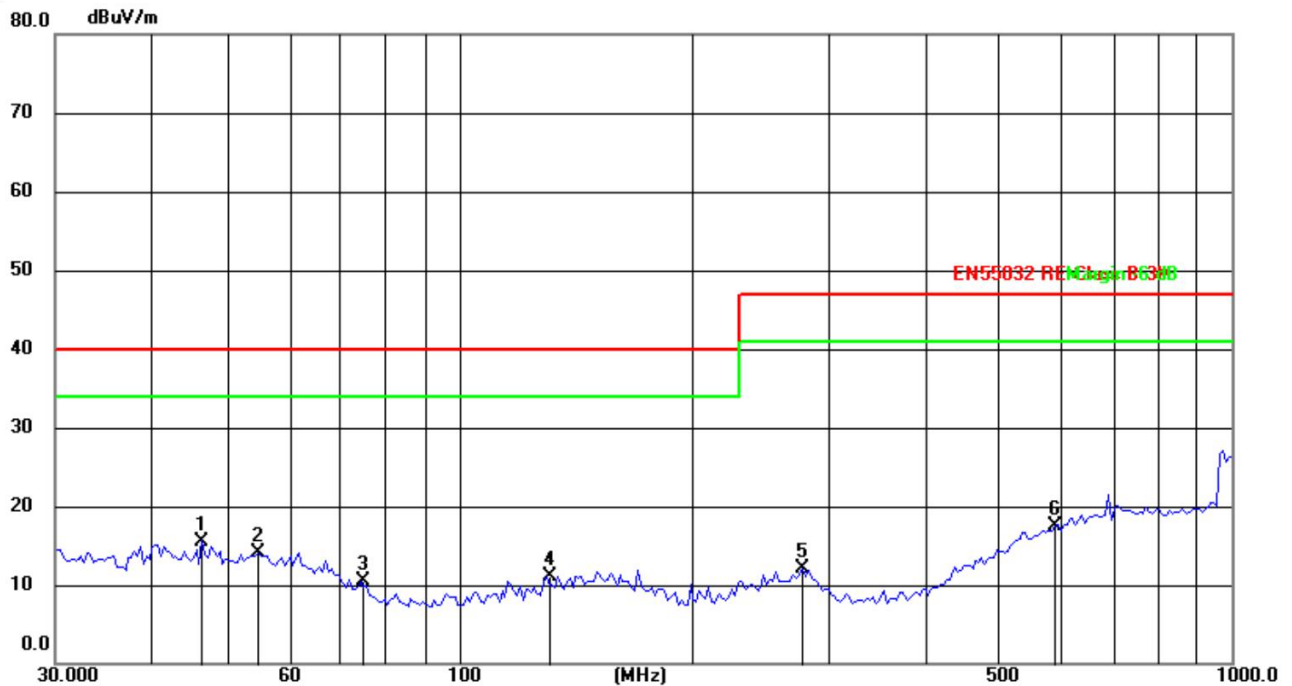
Above 1GHz:

- a. The Product was placed on the non-conductive turntable 0.8 m above the ground in a full anechoic chamber..
- b. Set the spectrum analyzer/receiver in Peak detector, Max Hold mode, and 1MHz RBW. Record the maximum field strength of all the pre-scan process in the full band when the antenna is varied in both horizontal and vertical, and the turntable is rotated from 0 to 360 degrees.
- c. For each frequency whose maximum record was higher or close to limit, measure its AV value: rotate the turntable from 0 to 360 degrees to find the degree where Product radiated the maximum emission, then set the test frequency analyzer/receiver to AV value and specified bandwidth with Maximum Hold Mode, and record the maximum value.



4.4 Test Results

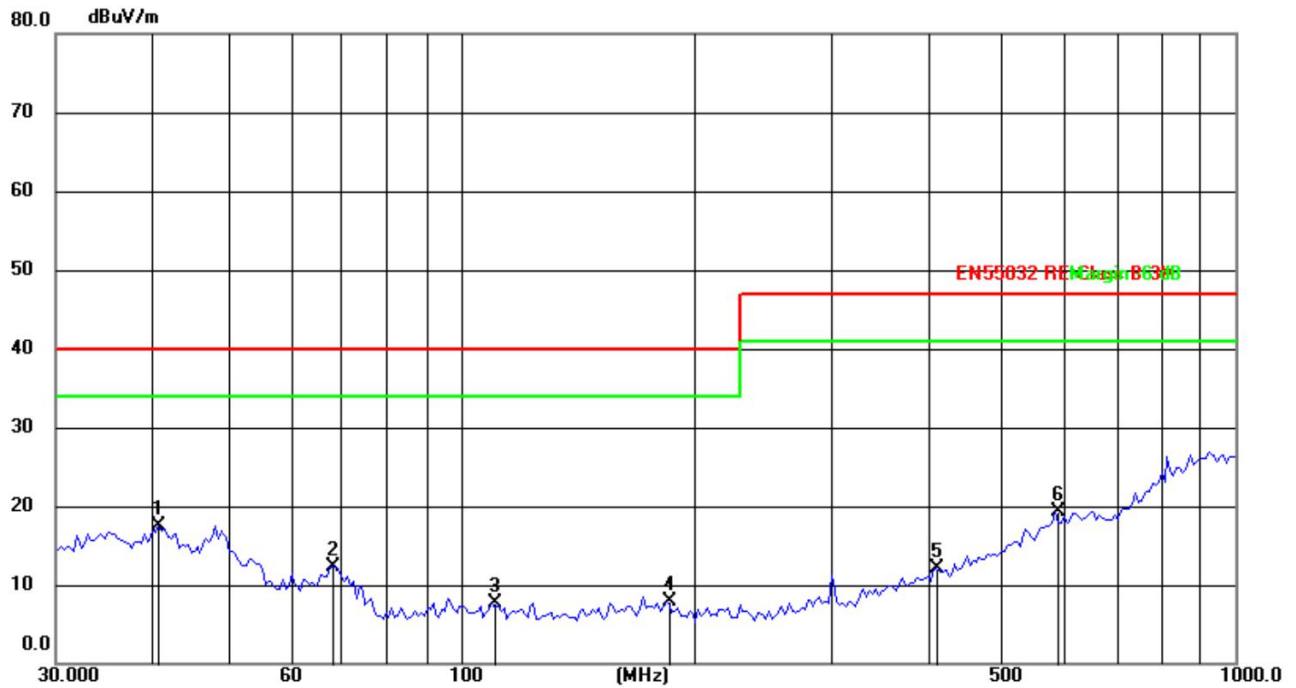
Radiated Emissions Test Data			
Temperature:	25.1℃	Relative Humidity:	51%
Pressure:	1009hPa	Phase :	Horizontal
Test Voltage :	DC 20V	Test Mode:	Working



No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	46.5030	29.68	-14.08	15.60	40.00	-24.40	QP				
2	54.9310	28.32	-14.14	14.18	40.00	-25.82	QP				
3	74.6569	28.61	-18.09	10.52	40.00	-29.48	QP				
4	129.6950	29.26	-18.07	11.19	40.00	-28.81	QP				
5	278.0668	28.34	-16.24	12.10	47.00	-34.90	QP				
6	590.9737	28.56	-11.07	17.49	47.00	-29.51	QP				



Radiated Emissions Test Data			
Temperature:	25.1℃	Relative Humidity:	51%
Pressure:	1009hPa	Phase :	Vertical
Test Voltage :	DC 20V	Test Mode:	Working

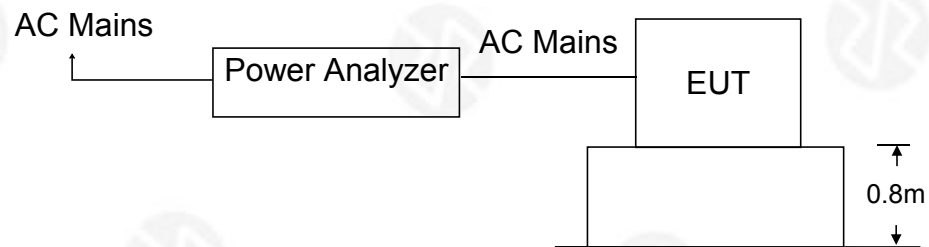


No.	Frequency (MHz)	Reading (dBuV)	Factor (dB/m)	Level (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Height (cm)	Azimuth (deg.)	P/F	Remark
1	40.7730	34.50	-16.90	17.60	40.00	-22.40	QP				
2	68.3908	31.51	-19.17	12.34	40.00	-27.66	QP				
3	109.7960	29.17	-21.40	7.77	40.00	-32.23	QP				
4	184.1667	28.81	-20.86	7.95	40.00	-32.05	QP				
5	408.9460	28.89	-16.73	12.16	47.00	-34.84	QP				
6	585.8157	29.59	-10.32	19.27	47.00	-27.73	QP				



5. HARMONIC CURRENT EMISSION TEST

5.1 Block Diagram of Test Setup



5.2 Test Standard

EN IEC 61000-3-2:2019/A1:2021

5.3 Operating Condition of EUT

- 5.3.1 Setup the EUT as shown in Section 6.1.
- 5.3.2 Turn on the power of all equipment.
- 5.3.3 Let the EUT work in test mode and test it.

5.4 Test Procedure

The power cord of the EUT is connected to the output of the test system. Turn on the power of the EUT and use the test system to test the harmonic current level.

5.5 Test Results

The EUT is powered by DC only. The test items is not applicable.



6. VOLTAGE FLUCTUATIONS & FLICKER TEST

6.1 Block Diagram of Test Setup

Same as Section 6.1.

6.2 Test Standard

EN 61000-3-3:2013+A1:2019

6.3 Operating Condition of EUT

Same as Section 5.3.. The power cord of the EUT is connected to the output of the test system. Turn on the power of the EUT and use the test system to test the harmonic current level.

Flicker Test Limit

Test items	Limits
Pst	1.0
dc	3.3%
dmax	4.0%
dt	Not exceed 3.3% for 500ms

6.4 Test Procedure

The power cord of the EUT is connected to the output of the test system. Turn on the power of the EUT and use the test system to test the harmonic current level.

6.5 Test Results

The EUT is powered by DC only. The test items is not applicable.



7. IMMUNITY TEST OF GENERAL THE PERFORMANCE CRITERIA

Product Standard	EN 55035:2017+A11:2020 clause 5
CRITERION A	The equipment shall continue to operate as intended without operator intervention. No degradation of performance, loss of function or change of operating state is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
CRITERION B	During the application of the disturbance, degradation of performance is allowed. However, no unintended change of actual operating state or stored data is allowed to persist after the test. After the test, the equipment shall continue to operate as intended without operator intervention; no degradation of performance or loss of function is allowed, below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level (or the permissible performance loss), or recovery time, is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
CRITERION C	Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. A reboot or re-start operation is allowed. Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

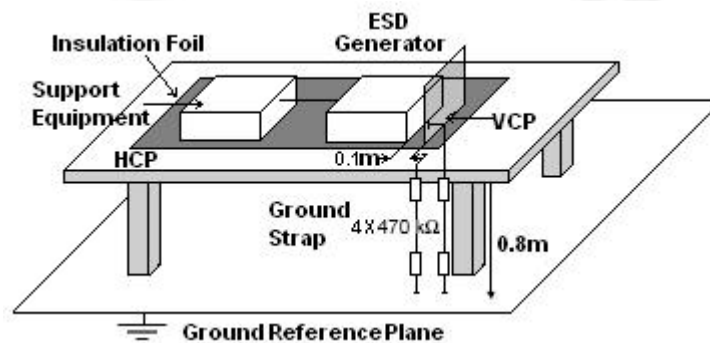


8. ELECTROSTATIC DISCHARGE (ESD)

8.1 Test Specification

Test Port	: Enclosure port
Discharge Impedance	: 330 ohm / 150 pF
Discharge Mode	: Single Discharge
Discharge Period	: one second between each discharge

8.2 Block Diagram of Test Setup



8.3 Test Procedure

- Electrostatic discharges were applied only to those points and surfaces of the Product that are accessible to users during normal operation.
- The test was performed with at least ten single discharges on the pre-selected points in the most sensitive polarity.
- The time interval between two successive single discharges was at least 1 second.
- The ESD generator was held perpendicularly to the surface to which the discharge was applied and the return cable was at least 0.2 meters from the Product.
- Contact discharges were applied to the non-insulating coating, with the pointed tip of the generator penetrating the coating and contacting the conducting substrate.
- Air discharges were applied with the round discharge tip of the discharge electrode approaching the Product as fast as possible (without causing mechanical damage) to touch the Product. After each discharge, the ESD generator was removed from the Product and re-triggered for a new single discharge. The test was repeated until all discharges were complete.
- At least ten single discharges (in the most sensitive polarity) were applied to the center of one vertical edge of the Vertical Coupling Plane in sufficiently different positions that the four faces of the Product were completely illuminated. The VCP (dimensions 0.5m x 0.5m) was placed vertically to and 0.1 meters from the Product.



8.4 Test Results

Discharge Method	Discharge Position	Voltage ($\pm$ kV)	Min. No. of Discharge per polarity (Each Point)	Required Level	Performance Criterion
Contact Discharge	Conductive Surfaces	4	10	B	A
	Indirect Discharge HCP	4	10	B	A
	Indirect Discharge VCP	4	10	B	A
Air Discharge	Slots, Apertures, and Insulating Surfaces	8	10	B	A
Note: N/A					



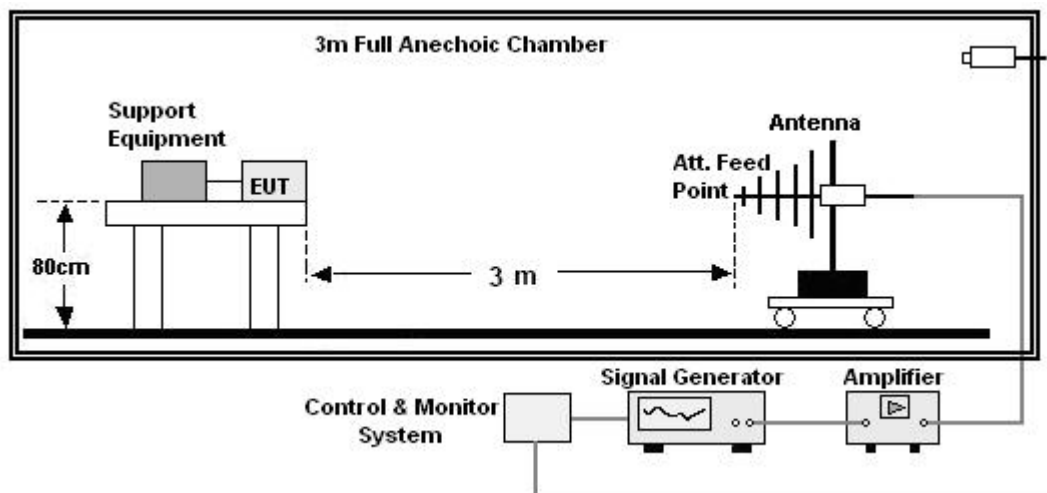
9. CONTINUOUS RF ELECTROMAGNETIC FIELD DISTURBANCES(RS)

9.1 Test Specification

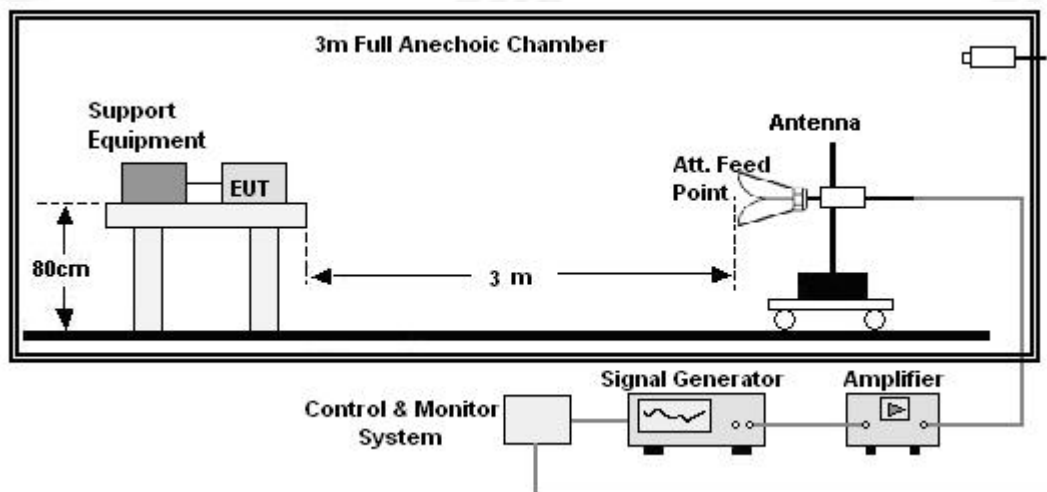
Test Port	: Enclosure port
Step Size	: 1%
Modulation	: 1kHz, 80% AM
Dwell Time	: 1 second
Polarization	: Horizontal & Vertical

9.2 Block Diagram of Test Setup

Below 1GHz:



Above 1GHz:





9.3 Test Procedure

- a. The testing was performed in a fully-anechoic chamber. The transmit antenna was located at a distance of 3 meters from the Product.
- b. The frequency range is swept from 80MHz to 1000MHz, 1800MHz, 2600MHz, 3500MHz, 5000MHz, with the signal 80% amplitude modulated with a 1 kHz sine wave, and the step size was 1%.
- c. The dwell time at each frequency shall not be less than the time necessary for the EUT to be exercised and to be able to respond, but should not exceed 5 s at each of the frequencies during the scan.
- d. The test was performed with the Product exposed to both vertically and horizontally polarized fields on each of the four sides.
- e. For Broadcast reception function: Group 2 not apply in this test.

9.4 Test Results

Frequency	Position	Field Strength (V/m)	Required Level	Performance Criterion
80 - 1000MHz, 1800MHz, 2600MHz, 3500MHz, 5000MHz	Front, Right, Back, Left	3	A	A
Note: N/A				



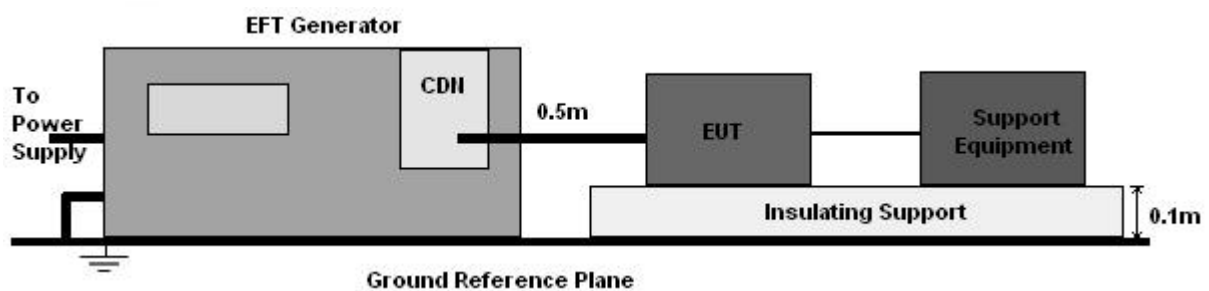
10. ELECTRICAL FAST TRANSIENTS/BURST (EFT)

10.1 Test Specification

Test Port	: input a.c. power port
Impulse Frequency	: 5 kHz
Impulse Wave-shape	: 5/50 ns
Burst Duration	: 15 ms
Burst Period	: 300 ms
Test Duration	: 2 minutes per polarity

10.2 Block Diagram of EUT Test Setup

For input a.c. power port:



10.3 Test Procedure

- The Product and support units were located on a non-conductive table above ground reference plane.
- A 0.5m-long power cord was attached to Product during the test.

10.4 Test Results

The EUT is powered by DC only. The test items is not applicable.

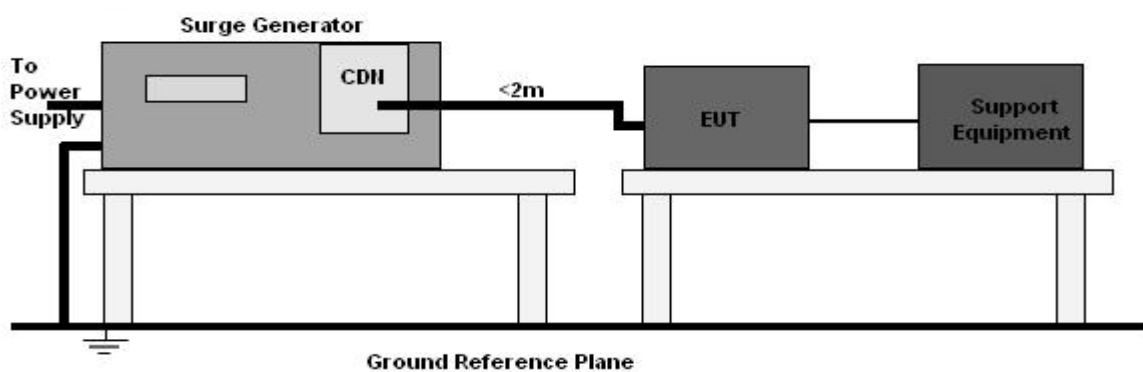


11. SURGES IMMUNITY TEST

11.1 Test Specification

Test Port	: input a.c. power port
Wave-Shape	: Open Circuit Voltage - 1.2 / 50 us Short Circuit Current - 8 / 20 us
Pulse Repetition Rate	: 1 pulse / min.
Phase Angle	: 0° / 90° / 180° / 270°
Test Events	: 5 pulses (positive & negative) for each polarity

11.2 Block Diagram of EUT Test Setup



11.3 Test Procedure

- The surge is to be applied to the Product power supply terminals via the capacitive coupling network. Decoupling networks are required in order to avoid possible adverse effects on equipment not under test that may be powered by the same lines, and to provide sufficient decoupling impedance to the surge wave.
- The power cord between the Product and the coupling/decoupling networks shall be 2 meters in length (or shorter). Interconnection line between the Product and the coupling/decoupling networks shall be 2 meters in length (or shorter).

11.4 Test Result

The EUT is powered by DC only. The test items is not applicable.



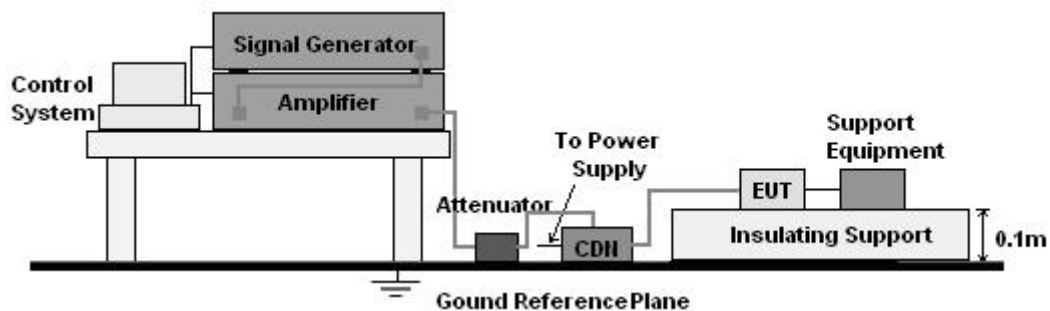
12. CONTINUOUS INDUCED RF DISTURBANCES (CS)

12.1 Test Specification

Test Port	: input a.c. power port
Step Size	: 1%
Modulation	: 1kHz, 80% AM
Dwell Time	: 1 second

12.2 Block Diagram of EUT Test Setup

For input a.c. power port:



12.3 Test Procedure

For input a.c. power port:

- The Product and support units were located at a ground reference plane with the interposition of a 0.1 m thickness insulating support and the CDN was located on GRP directly.
- The frequency range is swept from 150 kHz to 10MHz, 10MHz to 30MHz, 30MHz to 80MHz with the signal 80% amplitude modulated with a 1 kHz sine wave, and the step size was 1% of fundamental.
- The dwell time at each frequency shall be not less than the time necessary for the Product to be able to respond.

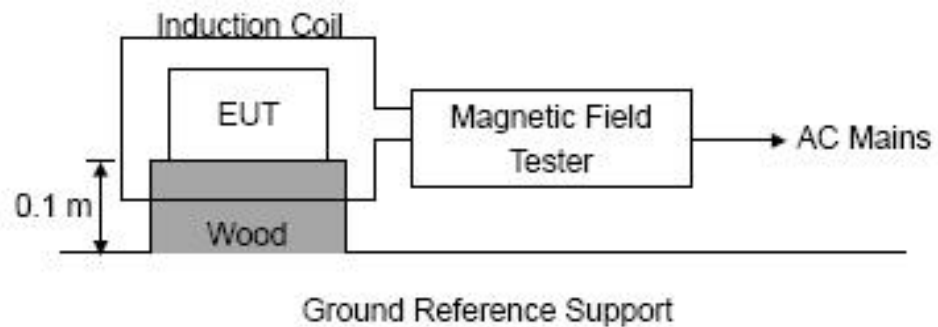
12.4 Test Result

The EUT is powered by DC only. The test items is not applicable.



13. MAGNETIC FIELD IMMUNITY TEST

13.1 Block Diagram of Test Setup



13.2 Test Standard

EN 55035:2017, EN61000-4-8:2010
Severity Level 1 at 1A/m

13.3 Severity Levels and Performance Criterion

13.3.1 Severity level

Level	Magnetic Field Strength A/m
1.	1
2.	3
3.	10
4.	30
5.	100
X.	Special

13.3.2 Performance criterion: B

- A. The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.
- B. After the test, the equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed. However, no change of



operating state or stored data is allowed to persist after the test.
If the minimum performance level (or the permissible performance loss) is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

- C. Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. Functions, and/or information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

13.4 EUT Configuration on Test

The configuration of EUT is listed in Section 2.9.

13.5 Operating Condition of EUT

Same as conducted emission test, which is listed in Section 2.9 except the test set up replaced as Section 12.1.

13.6 Test Procedure

The EUT shall be subjected to the test magnetic field by using the induction coil of standard dimensions (1m*1m) and shown in Section 10.1. The induction coil shall then be rotated by 90° in order to expose the EUT to the test field with different orientations.

13.7 Test Results

The EUT is powered by DC only. The test items is not applicable.

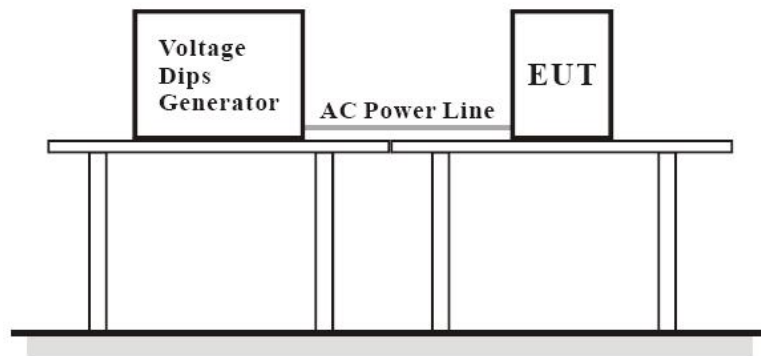


14. VOLTAGE DIPS AND INTERRUPTIONS (DIPS)

14.1 Test Specification

Test Port : input a.c. power port
Phase Angle : 0°, 180°
Test cycle : 3 times

14.2 Block Diagram of EUT Test Setup



14.3 Test Procedure

- The Product and support units were located on a non-conductive table above ground floor.
- Set the parameter of tests and then perform the test software of test simulator.
- Conditions changes to occur at 0 degree crossover point of the voltage waveform.

14.4 Test Result

The EUT is powered by DC only. The test items is not applicable.



15. EUT PHOTOGRAPHS

EUT Photo 1



EUT Photo 2





EUT Photo 3



EUT Photo 4





16. EUT TEST PHOTOGRAPHS

RE



***** END OF REPORT *****